

# **A Level Computer Science (9618)**

Topical Past Paper Worksheet

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## **Topic 2: Communication**

Subtopic 2.1: Networks including the Internet

**Concept 2.1.5: Networking Hardware (Switch, Router, NIC, WAP, etc.)**

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Total Questions: 12

**Mark Schemes begin on Page 8**

**Question 1 | Source: 9618\_s25\_qp\_12 (Q6.e)**

- 6 A company has multiple sites in different cities.  
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(e) Describe the role of a switch in a network.

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..... [3]

**Question 2 | Source: 9618\_s24\_qp\_12 (Q3.c.i)**

- 3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(c) Data transmitted on the internet passes through multiple different systems.

(i) Describe the role of routers in the transmission of data through the internet.

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..... [2]

**Question 3 | Source: 9618\_w23\_qp\_11 (Q2.a)**

- 2 A school has a Local Area Network (LAN).

- (a) The LAN connects to the internet using a router.

Describe the function of a router in a network.

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..... [3]

**Question 4 | Source: 9618\_w23\_qp\_11 (Q2.b)**

- 2 A school has a Local Area Network (LAN).

- (b) Complete the following table by writing the purpose of each of these other hardware devices used to support the LAN.

| Hardware device             | Purpose                 |
|-----------------------------|-------------------------|
| switch                      | .....<br>.....<br>..... |
| Wireless Access Point (WAP) | .....<br>.....<br>..... |
| bridge                      | .....<br>.....<br>..... |

[3]

**Question 5 | Source: 9618\_w23\_qp\_12 (Q7.b.ii)**

- 7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

- (ii) Explain how data is transmitted between the two laptops in the LAN.

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..... [2]

**Question 6 | Source: 9618\_w22\_qp\_12 (Q10.b.i)**

- 10 A Local Area Network (LAN) consists of four computers, one server and a switch.

The LAN uses a star topology.

- (b) A router is attached to one of the devices on the LAN shown in **part (a)** to connect the LAN to the internet.

- (i) Identify the device. Give a reason for your choice.

Device .....

Reason .....

.....

..... [2]

**Question 7 | Source: 9618\_w22\_qp\_12 (Q10.b.ii)**

- 10 A Local Area Network (LAN) consists of four computers, one server and a switch.

The LAN uses a star topology.

- (b) A router is attached to one of the devices on the LAN shown in **part (a)** to connect the LAN to the internet.

- (ii) Describe the role **and** function of the router in the network.

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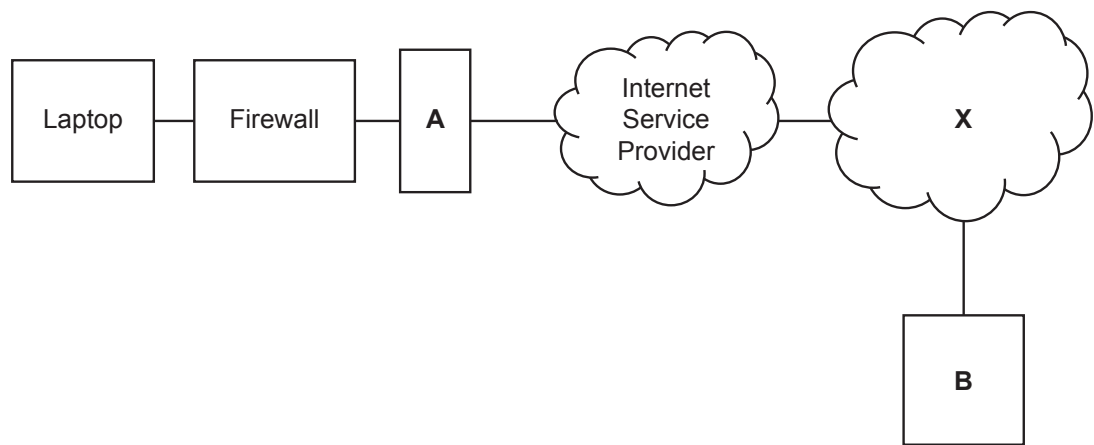
.....

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..... [3]

Question 8 | Source: 9618\_w22\_qp\_13 (Q7.a)

7 (a) The diagram shows the hardware and software used to retrieve files stored on the cloud.



Complete the following table by writing the answer for each row.

| Answer                                                                        |  |
|-------------------------------------------------------------------------------|--|
| The name of device <b>A</b> that allows the laptop to connect to the internet |  |
| A type of cloud, <b>X</b>                                                     |  |
| An example of an application, <b>B</b> , that can run on the cloud            |  |

[3]

Question 9 | Source: 9618\_w21\_qp\_11 (Q8.b)

8 A school is setting up a network within one of its buildings.

(b) One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

Identify **two** possible devices that can be used to physically connect the 30 computers to the rest of the network.

- 1 .....
- 2 .....

[2]

Question 10 | Source: 9618\_w21\_qp\_11 (Q8.c)

8 A school is setting up a network within one of its buildings.

- (c) The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

Describe the functions of a Wireless Network Interface Card.

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..... [4]

**Question 11 | Source: 9618\_w21\_qp\_12 (Q3.b.ii)**

- 3 Andy likes to play computer games.

- (b) Andy plays some of the computer games over the internet. He has several devices that connect wirelessly to the router in his house.

- (ii) The router has a wireless access point (WAP) to allow the devices to connect wirelessly.

Identify **three** functions of the router in Andy's network.

1 .....

.....

2 .....

.....

3 .....

..... [3]

**Question 12 | Source: 9618\_s21\_qp\_11 (Q4.c.i)**

- 4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

- (c) Melinda connects her laptop to the internet through her router.

- (i) Tick (✓) **one** box in each row to identify whether the task is performed by the router or not.

| <b>Task</b>                                                    | <b>Performed by router</b> | <b>Not performed by router</b> |
|----------------------------------------------------------------|----------------------------|--------------------------------|
| Receives packets from devices                                  |                            |                                |
| Finds the IP address of a Uniform Resource Locator (URL)       |                            |                                |
| Directs each packet to all devices attached to it              |                            |                                |
| Stores the IP and/or MAC address of all devices attached to it |                            |                                |

[2]

# MARK SCHEMES

## Mark Scheme for Question 1 | Source: 9618\_s25\_ms\_12 (Q6.e)

|      |                                                                                                                                                                                                                                                                                                                                                                                                |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(e) | <b>1 mark</b> each to <b>max 3</b><br><br>e.g. <ul style="list-style-type: none"> <li>• To store the MAC addresses of devices connected to it</li> <li>• To receive packets from devices</li> <li>• To forward packets directly to the intended recipient</li> <li>• To provide a central point of connection for the network</li> <li>• to enable connected devices to communicate</li> </ul> | <b>3</b> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

## Mark Scheme for Question 2 | Source: 9618\_s24\_ms\_12 (Q3.c.i)

|         |                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(c)(i) | <b>1 mark</b> each to <b>max 2</b> : <ul style="list-style-type: none"> <li>• Receives <b>packets</b> from internet</li> <li>• Analyses the destination IP address of each packet</li> <li>• Forwards the <b>packet</b> towards its destination</li> <li>• ... using the routing table</li> <li>• Maintains/updates the routing table</li> <li>• Finds the most efficient route to the destination</li> </ul> | <b>2</b> |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

## Mark Scheme for Question 3 | Source: 9618\_w23\_ms\_11 (Q2.a)

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <b>1 mark</b> for each bullet point ( <b>max 3</b> ) <ul style="list-style-type: none"> <li>• Receives packets from internet / external network</li> <li>• Implements a firewall</li> <li>• Analyses the destination IP address of each packet</li> <li>• Forwards the packet towards its destination // send packets onto local network or external network</li> <li>• ...using the routing table</li> <li>• Maintains / updates the routing</li> <li>• Allocates <u>private</u> IP addresses</li> <li>• Finds the most efficient route to the destination</li> <li>• Changes the packet format for transmission over the next network //</li> <li>• Network Address Translation (NAT): NAT is a technique used by routers to allow multiple devices in a private local area network (LAN) to share a single public IP address.</li> </ul> | <b>3</b> |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|



### Mark Scheme for Question 4 | Source: 9618\_w23\_ms\_11 (Q2.b)

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(b) | <p>Switch: <b>1 mark</b> for each bullet point (<b>max 1</b>)</p> <ul style="list-style-type: none"> <li>• To allow two or more devices to communicate with one another</li> <li>• To connect individual devices to each other</li> <li>• To receive transmissions and <b>forward</b> them to their destination</li> </ul> <p>Wireless Access Point (WAP): <b>1 mark</b> for each bullet point (<b>max 1</b>)</p> <ul style="list-style-type: none"> <li>• To allow connection of devices (to the central device) <b>using radio signals</b> / Wi-Fi</li> <li>• To allow the central device to send / receive <b>radio signals</b> / Wi-Fi signals</li> <li>• To allow wireless enabled devices to connect to a wired network</li> </ul> <p>Bridge: <b>1 mark</b> for each bullet point (<b>max 1</b>)</p> <ul style="list-style-type: none"> <li>• To connect two LANs / segments with the <b>same protocol</b></li> <li>• To transmit data between two networks with the <b>same protocol</b></li> </ul> | <b>3</b> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 5 | Source: 9618\_w23\_ms\_12 (Q7.b.ii)

|          |                                                                                                                                                                                                                                                                                                                                                                                                     |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b)(ii) | <p><b>1 mark</b> for each bullet point (<b>max 2</b>)</p> <ul style="list-style-type: none"> <li>• The data from the sending laptop is transmitted to the router</li> <li>• The data has address of recipient</li> <li>• The router determines recipient's destination address</li> <li>• ... by using a routing table</li> <li>• The router transmits data directly / only to recipient</li> </ul> | <b>2</b> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 6 | Source: 9618\_w22\_ms\_12 (Q10.b.i)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(b)(i) | <p><b>1 mark</b> for the device. <b>1 mark</b> for corresponding reason.</p> <ul style="list-style-type: none"> <li>• Device: Server</li> <li>• Reason: Server processes the requests and authorises traffic // firewall software on the server authorises traffic // server acts as the proxy</li> <li>• Device: Switch</li> <li>• Reason: Switch is connected to all the computers // to share access to the router on the network</li> </ul> | <b>2</b> |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 7 | Source: 9618\_w22\_ms\_12 (Q10.b.ii)

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(b)(ii) | <b>1 mark</b> for each bullet point ( <b>max 3</b> ): <ul style="list-style-type: none"> <li>• receive packets from devices / internet</li> <li>• find destination of packets using the IP address</li> <li>• forward packets to <b>the destination</b></li> <li>• assign <b>private</b> IP addresses to devices on LAN</li> <li>• store/update/maintain a routing table</li> <li>• find most efficient <b>path</b> to destination</li> <li>• maintain table of MAC and IP addresses</li> <li>• provides the LAN with a public IP address</li> <li>• acts as a gateway</li> <li>• performs protocol conversion</li> <li>• acts as a firewall</li> </ul> | <b>3</b> |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 8 | Source: 9618\_w22\_ms\_13 (Q7.a)

|                                                                        |                                                                         |          |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|----------|
| 7(a)                                                                   | <b>1 mark</b> for each correct answer:                                  | <b>3</b> |
|                                                                        | <b>Answer</b>                                                           |          |
| The name of device A that allows the laptop to connect to the internet | Router                                                                  |          |
| A type of cloud, X                                                     | Public (cloud)                                                          |          |
| An example of an application, B, that can run on the cloud,            | Email / Graphics / Word processor / Spreadsheet / Game / Database, etc. |          |

### Mark Scheme for Question 9 | Source: 9618\_w21\_ms\_11 (Q8.b)

|      |                                                                                                                    |          |
|------|--------------------------------------------------------------------------------------------------------------------|----------|
| 8(b) | <b>1 mark</b> per item <ul style="list-style-type: none"> <li>• router</li> <li>• switch</li> <li>• hub</li> </ul> | <b>2</b> |
|------|--------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 10 | Source: 9618\_w21\_ms\_11 (Q8.c)

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8(c) | <b>1 mark</b> per bullet point to <b>max 4</b> <ul style="list-style-type: none"> <li>• Provide interface to wireless network</li> <li>• ... as an antenna</li> <li>• Receives analogue radio waves</li> <li>• ... convert them to digital / binary</li> <li>• Checks incoming transmissions for correct MAC / IP address</li> <li>• ... ignore transmissions not intended for it</li> <li>• Encrypts / encodes the data</li> <li>• Decrypts / decodes the data</li> <li>• Takes digital/binary input and converts to analogue waves</li> <li>• ... sends the radio waves via the antenna</li> </ul> | <b>4</b> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 11 | Source: 9618\_w21\_ms\_12 (Q3.b.ii)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b)(ii) | <b>1 mark</b> for each correct function to <b>max 3</b> <ul style="list-style-type: none"> <li>• To receive packets from devices or the Internet</li> <li>• To forward / route packets to the destination</li> <li>• To find the destination of the packet</li> <li>• To assign / allocate private IP addresses to devices on LAN</li> <li>• To store / update / maintain a routing table</li> <li>• To find the most efficient path to the destination</li> <li>• To maintain a table of MAC and IP addresses</li> </ul> | <b>3</b> |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

### Mark Scheme for Question 12 | Source: 9618\_s21\_ms\_11 (Q4.c.i)

| 4(c)(i)                                                        | <b>1 mark</b> for first 2 ticks, <b>1 mark</b> for last 2 (shaded) <table border="1"> <thead> <tr> <th>Task</th><th>Performed by router</th><th>Not performed by router</th></tr> </thead> <tbody> <tr> <td>Receives packets from devices</td><td>✓</td><td></td></tr> <tr> <td>Finds the IP address of a Uniform Resource Locator (URL)</td><td></td><td>✓</td></tr> <tr> <td>Directs each packet to all devices attached to it</td><td></td><td>✓</td></tr> <tr> <td>Stores the IP and/or MAC address of all devices attached to it</td><td>✓</td><td></td></tr> </tbody> </table> | Task                    | Performed by router | Not performed by router | Receives packets from devices | ✓ |  | Finds the IP address of a Uniform Resource Locator (URL) |  | ✓ | Directs each packet to all devices attached to it |  | ✓ | Stores the IP and/or MAC address of all devices attached to it | ✓ |  | <b>2</b> |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------|-------------------------|-------------------------------|---|--|----------------------------------------------------------|--|---|---------------------------------------------------|--|---|----------------------------------------------------------------|---|--|----------|
| Task                                                           | Performed by router                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Not performed by router |                     |                         |                               |   |  |                                                          |  |   |                                                   |  |   |                                                                |   |  |          |
| Receives packets from devices                                  | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |                     |                         |                               |   |  |                                                          |  |   |                                                   |  |   |                                                                |   |  |          |
| Finds the IP address of a Uniform Resource Locator (URL)       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ✓                       |                     |                         |                               |   |  |                                                          |  |   |                                                   |  |   |                                                                |   |  |          |
| Directs each packet to all devices attached to it              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ✓                       |                     |                         |                               |   |  |                                                          |  |   |                                                   |  |   |                                                                |   |  |          |
| Stores the IP and/or MAC address of all devices attached to it | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |                     |                         |                               |   |  |                                                          |  |   |                                                   |  |   |                                                                |   |  |          |